CanonDressGS: Reference-Controlled Garment Endpoint Selection in Canonical Gaussian Space

Anonymous submission

Abstract

Animatable Gaussian avatars are usually tied to the garment observed during capture. Directly interpolating garment residuals appears to offer control, but it can move Gaussian geometry into invalid intermediate supports; a controlled decomposition attributes a dominant effect of 0.995941 to geometry and finds it sufficient and necessary in all ten garment pairs. We introduce CanonDressGS, a reference-controlled method for a fixed avatar and a closed bank of seen garments. Offline multi-view optimization constructs valid canonical Teacher Endpoints and a Teacher-derived coordinate system. At inference, frozen reference features drive a 3,076-parameter endpoint head, whose raw coefficient is snapped to the nearest valid endpoint before frozen deformation and rendering. On subject02 with five seen garments, the method attains 100% clean endpoint-selection accuracy with no observed identity contamination, while exposing its clean functional equivalence to hard lookup. A separate Dual-Support extension preserves two valid geometry supports for oracle- or user-specified composition and reduces pair-average LPIPS from 0.085303 to 0.056985 over all ten pairs. Negative headroom and leave-one-garment-out analyses show that the global affine endpoint space does not support reliable held-out-garment adaptation. The current evidence is therefore limited to one identity, seen-garment references, and closed-wardrobe endpoint realization.

Introduction

Animatable Gaussian avatars provide efficient pose-driven rendering, but their canonical Gaussian state commonly entangles identity with the garment present during capture. Changing the garment is therefore not only an appearance-editing problem: it changes the spatial support, scale, rotation, opacity, and appearance carried by a large set of canonical Gaussians. A garment-control mechanism must preserve the frozen identity, deformation model, and renderer while replacing this coupled state.

The tempting construction is to interpolate residual parameters between two optimized garments. Our causal analysis shows why this is unsafe. Interpolated geometry can occupy unsupported locations and scales even when endpoint states are individually valid. Across all ten unordered pairs in the five-garment study, geometry has a global main-effect magnitude of 0.995941 and is both sufficient and necessary for the observed invalid intermediate states in every pair. Appearance or visibility changes alone do not explain the failures.

We consequently formulate garment control as endpoint realization rather than arbitrary intermediate-state generation. For a fixed avatar, we optimize one valid canonical Teacher Endpoint per seen garment and express the endpoint bank in an explicit affine coordinate system. Given one or more reference RGB images and masks, a small head predicts an endpoint-space coefficient. The prediction is not rendered directly: nearest-endpoint snapping deterministically realizes one registered valid support and then reuses the frozen deformation and renderer.

This endpoint view also clarifies a separate mixed-composition extension. Dual-Support keeps two optimized geometry supports separate and combines their effective contributions, avoiding direct geometry interpolation. The pair and mixture weight are currently oracle- or user-specified; automatic mixed-reference routing is outside the demonstrated method.

Our formal evaluation uses subject02, five seen garments (O01, O02, O03, O04, and O08), and a closed reference bank. It does not test a second identity or unseen garments. Within this scope, clean endpoint selection is exact and identity-safe, whereas headroom and leave-one-garment-out analyses expose the limits of the global affine coordinate system.

Our contributions are:

- a construction of canonical garment endpoints and an explicit Teacher-derived endpoint coordinate system for a frozen animatable Gaussian avatar;
- a controlled causal decomposition showing that Gaussian geometry interpolation is the dominant source of invalid intermediate garment states; and
- a lightweight reference-controlled endpoint-realization framework that preserves identity by snapping predictions to valid garment endpoints.

We additionally study Dual-Support as a geometry-safe extension for oracle- or user-specified composition.

Related Work

Animatable Human and Gaussian Avatars

Animatable human models factor a canonical representation from pose-driven deformation and rendering. Gaussian avatars make this factorization efficient by deforming and rasterizing explicit primitives (Li et al. 2024; Qian et al. 2024; Kocabas et al. 2024), but the canonical state still reflects the captured clothing. CanonDressGS freezes the avatar, deformation, skinning, and renderer; it addresses only the canonical garment state needed to use that existing avatar.

Garment-Aware Human Modeling and Editing

Garment-aware reconstruction and editing methods model clothing as geometry, appearance, deformation, or a combination of these factors (Ma et al. 2020; Patel, Liao, and Pons-Moll 2020; Kim et al. 2024). Our setting is narrower than garment generation or virtual try-on: every realizable garment is first optimized as a valid endpoint on one fixed avatar. This restriction permits an explicit test of whether interpolation preserves Gaussian support and leads to the endpoint-selection formulation.

Reference-Conditioned State Control

Image-based virtual try-on has used garment reference images to synthesize edited person images (Neuberger et al. 2020). Our setting instead predicts a state in a fixed canonical avatar. The predictor uses frozen per-reference features and permutation-invariant set aggregation, but its deployed output is discrete: a raw coefficient is mapped to the nearest member of a registered endpoint bank. This distinction separates the demonstrated method from continuous garment regression and automatic mixed-reference composition.

Method

Problem Setting and Scope

Let a fixed identity be represented by a frozen animatable Gaussian avatar with canonical base state G_{base}, pose-driven deformation and linear blend skinning (LBS), and a Gaussian renderer. Let $\mathcal{W} = \{\text{O01}, \text{O02}, \text{O03}, \text{O04}, \text{O08}\}$ be a closed bank of seen garments. Given one or more reference RGB images, clothing masks, foreground masks, and validity flags for a target $g \in \mathcal{W}$, the task is to realize the corresponding valid canonical garment endpoint. The task does not include unseen-garment synthesis, cross-person transfer, or arbitrary interpolation.

Frozen Animatable Gaussian Avatar

The base state G_{base} contains the canonical Gaussian support and attributes required by the existing avatar. CanonDressGS does not retrain this state or its deformation stack. Canonical garment residuals are applied before the frozen pose-dependent deformation, LBS, and rendering operations. This ordering preserves the avatar's identity and motion model while allowing the canonical garment state to change.

Canonical Teacher Endpoint Construction

For each $g \in \mathcal{W}$, frozen-avatar multi-view optimization produces a canonical residual ΔG_g^* and endpoint

$$G_g^* = G_{\text{base}} + \Delta G_g^*. \tag{1}$$

Each G_g^* is an optimized valid endpoint under the frozen avatar and renderer. It is not a ground-truth garment mesh. The endpoint bank is constructed offline and remains fixed during reference-conditioned training and inference.

Explicit Endpoint Coordinate System

We vectorize the five Teacher residuals, compute their mean

$$\mu = \frac{1}{|\mathcal{W}|} \sum_{g \in \mathcal{W}} \Delta G_g^*, \tag{2}$$

and obtain $B \in \mathbb{R}^{D \times r}$ from deterministic mean-centered SVD. With five endpoints, the maximum affine rank is $r = 4$. The coefficient of endpoint g is

$$c_g = B^\top (\Delta G_g^* - \mu), \qquad \widehat{\Delta G}(c) = \mu + Bc. \tag{3}$$

This basis is an endpoint coordinate system over the registered wardrobe, not a semantic or generalizable garment manifold.

Reference Feature Aggregation

For each valid reference i, frozen F2 processing yields $f_i \in \mathbb{R}^{256}$ using the registered clothing-weighted mean and clothing-masked max contract. A permutation-invariant set representation concatenates validity-aware mean and max aggregation,

$$h = \left[\frac{1}{N_v} \sum_{i:v_i=1} f_i \ ; \ \max_{i:v_i=1} f_i \right] \in \mathbb{R}^{512}, \tag{4}$$

where v_i is the reference-validity indicator and $N_v = \sum_i v_i$. An empty valid set is handled by the safe fallback described below, so Eq. 4 is evaluated only for $N_v > 0$.

Endpoint Head and Snapping

The trainable head is $\text{Linear}_{512 \to 4}(\text{LayerNorm}(h))$ and has 3,076 parameters. Let $m, s \in \mathbb{R}^4$ be the registered mean and standard deviation of the endpoint coefficients and $z_g = (c_g - m) \oslash s$ the standardized endpoint. The head predicts a raw standardized coefficient $\widetilde{z}$. CanonDressGS selects

$$\widehat{g} = \arg\min_{g \in \mathcal{W}} \|\widetilde{z} - z_g\|_2^2, \qquad c_{\text{realized}} = c_{\widehat{g}}. \tag{5}$$

The raw prediction and realized snapped coefficient are therefore distinct. Only $\widehat{\Delta G}(c_{\text{realized}})$ is passed to the frozen deformation and renderer. Snapping prevents unsupported interpolated geometry, restricts realization to optimized supports, and converts approximate coefficient prediction into stable endpoint selection. If no reference is valid, the method abstains and renders G_{base}.

Figure 2 summarizes the offline endpoint construction and the reference-conditioned realization path, including the distinction between raw prediction and the snapped endpoint actually rendered.

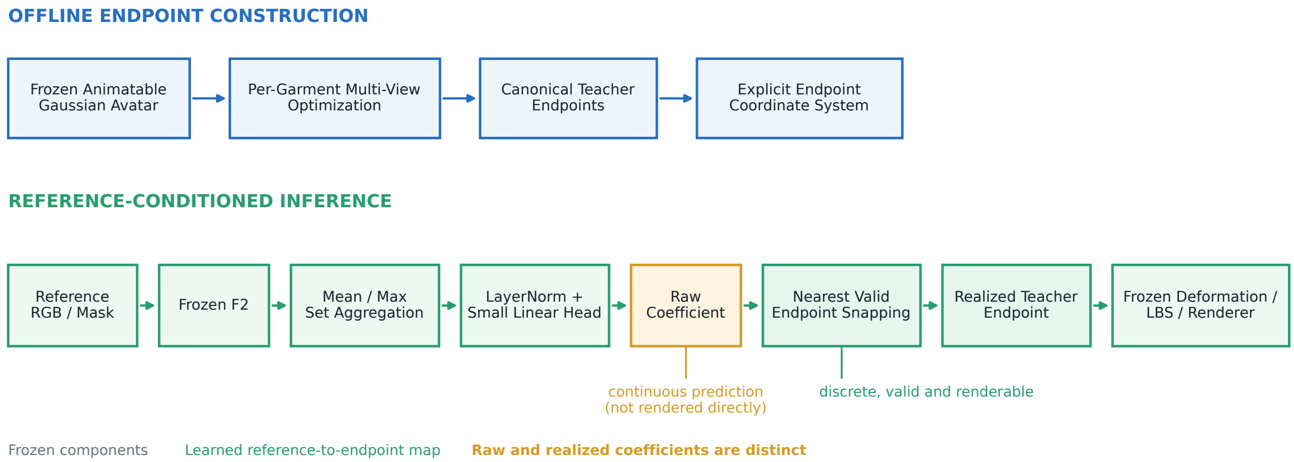

Figure 2: CanonDressGS method overview for the fixed subject02 identity and five-seen-garment closed wardrobe. Offline, per-garment multi-view optimization constructs canonical Teacher Endpoints and an explicit endpoint coordinate system for a frozen animatable Gaussian avatar. At inference, frozen reference features are aggregated by mean and max, and a small normalized linear head predicts a raw coefficient. The raw coefficient is not rendered directly: nearest-valid-endpoint snapping realizes a registered Teacher Endpoint before frozen deformation, LBS, and rendering.

Geometry-Safe Dual-Support Extension

For oracle- or user-specified endpoints $g_a, g_b \in \mathcal{W}$ and weight α, Dual-Support retains $G^*_{g_a}$ and $G^*_{g_b}$ as separate Gaussian supports. Geometry is not interpolated. Instead, the two valid supports are rendered with combined opacity or effective contribution according to α. This extension can compose two registered garments without forcing their geometry into an invalid midpoint, but it approximately doubles active Gaussians. Predicting the pair or weight from mixed references is unresolved and is not part of the default single-endpoint path.

Training Objectives

Training uses only the standardized coefficient target. For a four-dimensional prediction $\widetilde{z}$ and target z_g, the objective is

$$\mathcal{L}_{\text{coef}} = \frac{1}{4}\sum_{j=1}^{4} \mathrm{SmoothL1}(\widetilde{z}_j - z_{g,j}; \beta = 1), \tag{6}$$

with the usual quadratic branch for absolute error below one and linear branch otherwise. There is no render loss, classification loss, pair-selection loss, or mixture-controller loss in the frozen endpoint method.

Experiments

Experimental Setup

We evaluate subject02 with the closed garment bank O01, O02, O03, O04, and O08. The avatar, Teacher Endpoint bank, rank-4 coordinate system, F2 features, deformation, LBS, and renderer are frozen. Four condition-fold rotations use two folds for training, one for calibration, and one for testing. Record identifiers and logical query hashes are disjoint across these partitions, but exact reference assets can recur; this is a closed-wardrobe condition split, not a novel-view or unseen-reference protocol. The formal execution contains 24 optimizer runs and 7,200 optimizer steps across the two trainable predictor families. We report unique-query endpoint identification, per-garment precision and recall, exact endpoint realization, image metrics, and registered visual safety grades.

Baselines

The matched table includes the Base Avatar, the non-deployable Teacher Endpoint and Outfit-ID Oracle references, a trained Reference Classifier Lookup, a fixed Nearest-Centroid Lookup, and CanonDressGS Endpoint. The Linear Coefficient Predictor shares CanonDressGS training but renders its raw coefficient directly and is used only to analyze the role of snapping. A historical Direct Residual Decoder has unmatched exploratory provenance and is excluded from matched comparisons.

Geometry Causal Attribution

We intervene independently on Gaussian geometry, visibility, and appearance for all ten unordered garment pairs. Geometry has a global main-effect magnitude of 0.995941 and satisfies the registered sufficiency and necessity criteria in 10/10 pairs. Visibility and appearance satisfy neither criterion in any pair. Figure 3 visualizes the decomposition and all-pair criteria. The result motivates valid-support realization rather than direct residual interpolation.

Closed-Wardrobe Endpoint Selection

Table 1 reports the unique-query clean condition. CanonDressGS selects all 60 endpoints correctly across four rota-

Method	Top-1	Macro P	Macro R	Exact	Identity	Severe	Params.
Base Avatar	–	–	–	–	–	–	0
Teacher Endpoint	1.000	1.000	1.000	–	–	–	0
Outfit-ID Oracle	1.000	1.000	1.000	1.000	–	–	0
Reference Classifier Lookup	1.000	1.000	1.000	1.000	–	–	3,589
Nearest-Centroid Lookup	1.000	1.000	1.000	1.000	–	–	0
CanonDressGS Endpoint	1.000	1.000	1.000	1.000	0	0	3,076

Table 1: Subject02 closed-wardrobe endpoint selection on 60 unique clean queries. Precision and recall are macro averages over O01, O02, O03, O04, and O08. Dashes denote metrics that are not defined for a non-predictive reference. Teacher Endpoint and Outfit-ID Oracle are evaluation references, not deployable controllers.

tions: clean top-1, macro precision, macro recall, and endpoint exact match are all 1.000. Every garment has precision and recall 1.000, and every rotation has top-1 1.000. No identity or component contamination and no severe wrong outfit are observed. The Teacher image metrics establish parity, not a quality improvement over the Teacher. Reference Classifier Lookup, Nearest-Centroid Lookup, Outfit-ID Oracle, and CanonDressGS select the same clean endpoints; Figure 5 discloses this functional equivalence rather than claiming superiority.

Robustness and Safe Fallback

Input perturbations reveal prediction sensitivity without producing interpolated geometry artifacts. CanonDressGS clean top-1 drops to 0.350 under mild blur and reaches 0.950 with one reference; its registered endpoint-flip rates are 0.650 and 0.050, respectively. Assignment permutation, mask dilation, mask erosion, and ordinary reference dropout leave endpoint selection unchanged. Across 80 complete-reference-dropout cases, the empty-set guard abstains and returns the Base Avatar. Figure 6(C) summarizes these registered rates. Thus an incorrect endpoint is a prediction failure, while the snapping and abstention rules prevent an unsupported intermediate geometry from being rendered.

Dual-Support Analysis

The all-pair evaluation covers all ten unordered pairs, both directions, three mixture weights, and four views, without pair or view selection. Relative to full residual interpolation, Dual-Support improves pair-average LPIPS from 0.085303 to 0.056985, silhouette IoU from 0.710052 to 0.725290, and boundary F-score from 0.300263 to 0.379103. Severe-artifact counts decrease for all ten pairs and identity-contamination grade remains zero. The result is not artifact-free: O01–O03 and O02–O03 retain severe patch, cloud, mottle, or full-body contamination at the central mixture. Figure 4 reports the complete all-pair LPIPS profile, aggregate metrics, visual audit, and cost; Table 2 gives the corresponding tradeoff.

Endpoint-Space Headroom

Test-time coefficient refinement provides no additional headroom. Teacher Endpoint LPIPS is 0.038768, whereas the refined coefficient obtains 0.038982; the Teacher-relative macro gain is negative and 0/5 garments meet the improvement criterion. Figure 6(A) reports this boundary. A full-residual optimizer is worse (LPIPS 0.118421) and produces severe patch, cloud, and mottle artifacts. We therefore omit test-time refinement from the inference pipeline.

Leave-One-Garment-Out Capacity Analysis

For each held-out garment, the remaining four centered endpoints span at most a rank-3 affine basis. Offline oracle projection fails its capacity criterion in all five folds, and deployable low-dimensional adaptation does not reliably improve over nearest hard lookup: macro LPIPS is 0.113551 versus 0.113437. Increasing the reference budget is also negative. Figure 6(B) identifies this as a representational boundary rather than an identity-safety failure: identity and component contamination remain zero. We retain the full five-garment rank-4 basis as a closed-wardrobe coordinate system, not as evidence of held-out-garment adaptation.

Limitations

The formal evaluation is limited to subject02, so it does not establish cross-identity generalization. The wardrobe is closed and contains five seen garments; unseen garments and open-world synthesis are unsupported. Although the head predicts a coefficient, the deployed method is endpoint selection: under clean references it is functionally equivalent to the registered hard-lookup baselines, and its raw coefficient is not evidence of accurate continuous control.

The global affine coordinate system also has limited scope. Test-time coefficient refinement yields no improvement, and a four-garment rank-3 basis cannot represent the held-out fifth garment reliably. Richer localized representations would be required before claiming held-out adaptation. Dual-Support does not resolve automatic mixed-reference control because its endpoint pair and weight are oracle- or user-specified. It uses 1.999817 times as many active Gaussians and 1.625864 times the render time of a single hard support, and two pairs retain severe residual artifacts.

Finally, endpoint prediction is sensitive to reference quality: mild blur reduces cleanly registered top-1 from 1.000 to 0.350, and a single reference gives 0.950. Complete reference dropout is safe because the system abstains and returns the Base Avatar, but that behavior is a safety fallback rather than garment recovery.

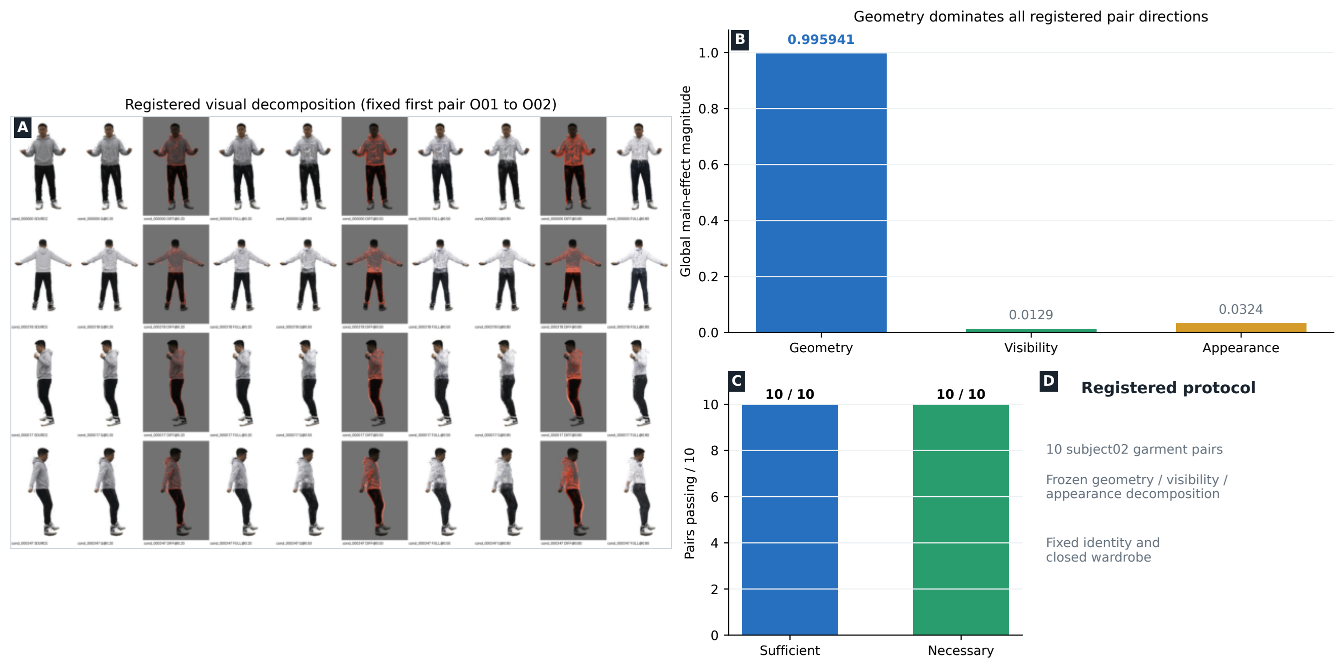

Figure 3: Geometry interpolation causal attribution over all 10 subject02 garment pairs under the frozen geometry/visibility/appearance decomposition, fixed identity, and closed wardrobe. (A) A registered O01-to-O02 visual decomposition; this fixed first pair is illustrative rather than quality-selected. (B) Geometry has global main-effect magnitude 0.995941, compared with 0.0129 for visibility and 0.0324 for appearance. (C) Geometry is sufficient and necessary in 10/10 pairs. (D) Scope of the registered protocol.

Method	LPIPS	IoU	B-F	Active	Time
Full interpolation	0.085303	0.710052	0.300263	1.000000	–
Dual-Support	0.056985	0.725290	0.379103	1.999817	1.625864

Table 2: Geometry-safe Dual-Support analysis over all ten garment pairs. Active-count ratios are relative to one hard geometry support; the reused full-interpolation baseline has no registered timing measurement. Lower LPIPS is better; higher IoU and boundary F-score are better.

Conclusion

CanonDressGS reframes garment control for a frozen animatable Gaussian avatar as reference-controlled realization of valid canonical endpoints. A causal decomposition identifies geometry interpolation as the dominant source of invalid intermediate states, motivating nearest-endpoint snapping instead of direct residual interpolation. Within a fixed-identity, five-garment closed wardrobe, the resulting endpoint method is exact on clean references and preserves identity. Dual-Support offers a separate geometry-safe mechanism for oracle- or user-specified mixed composition while retaining two valid supports. Headroom and leave-one-garment-out analyses delimit the method: the current affine space is useful as an endpoint coordinate system but not as a held-out-garment adaptation space. A central future direction is developing richer localized garment representations for open-set or held-out garment adaptation.

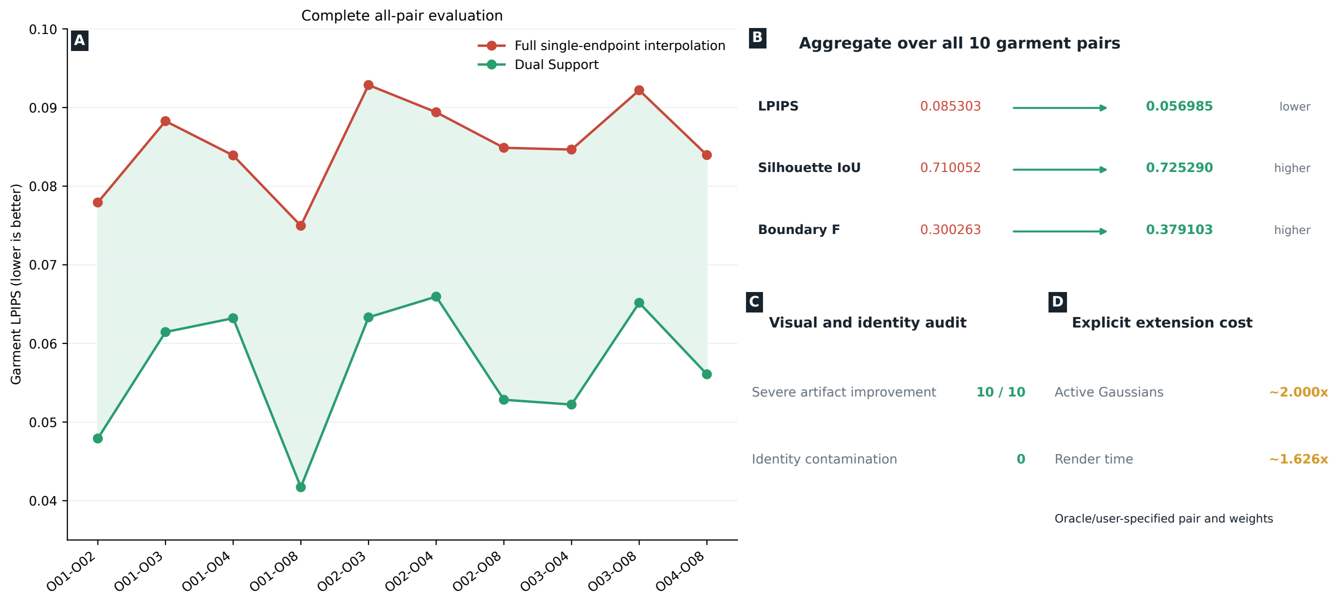

Figure 4: Dual-Support all-pair evaluation in the fixed-identity subject02 five-seen-garment closed wardrobe, using an oracle- or user-specified endpoint pair and weight. (A) LPIPS for all 10 unordered garment pairs. (B) Aggregate LPIPS improves from 0.085303 to 0.056985, silhouette IoU from 0.710052 to 0.725290, and Boundary F from 0.300263 to 0.379103. (C) Severe artifacts improve in 10/10 pairs with zero identity contamination. (D) The extension uses approximately twice the active Gaussians and 1.626 times the render time. Pair and weight are specified externally; no routing predictor is evaluated.

Analysis	Evidence	Method implication
Test-time headroom	LPIPS 0.038768 $\rightarrow$ 0.038982; 0/5 improve	Keep the frozen Teacher Endpoint; omit refinement.
Full residual	LPIPS 0.118421; severe artifacts	Diagnostic only; do not enter the main pipeline.
Leave-one-out basis	Rank 3; 0/5 pass capacity	The affine basis is not a held-out adaptation space.
Hard lookup	0.113437 vs. 0.113551 low-dimensional LPIPS	Retain as a descriptive closed-wardrobe relation.

Table 3: Endpoint-space scope analysis. Headroom and leave-one-garment-out results are negative diagnostics that delimit the method; they are not deployment improvements.

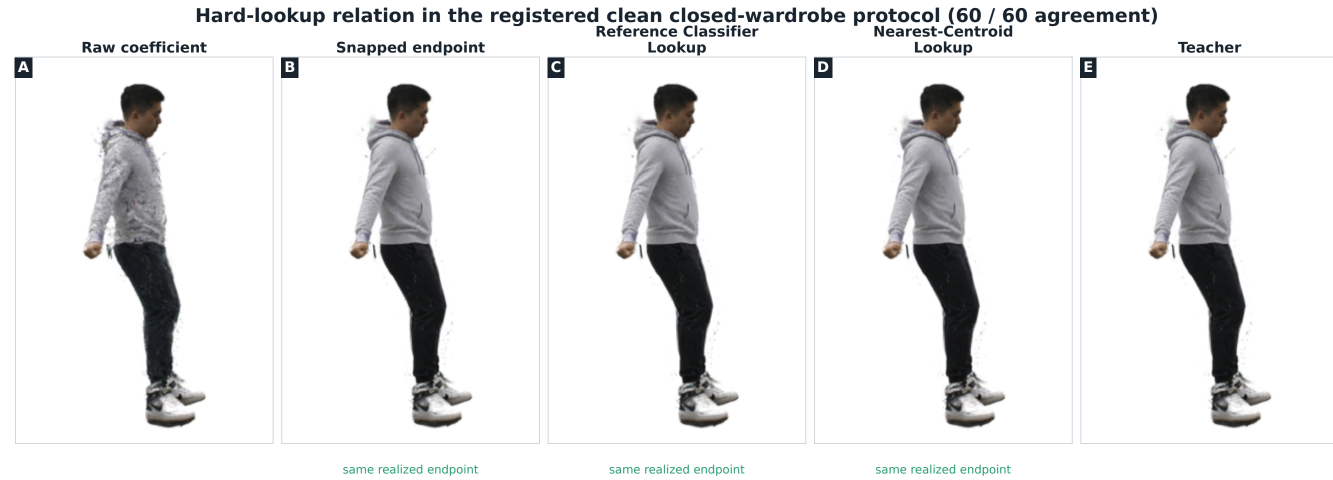

Figure 5: Hard-lookup relation in the registered clean fixed-identity subject02, five-seen-garment closed-wardrobe protocol (60 unique queries). The raw continuous coefficient can produce an unsupported render, whereas CanonDressGS snapping, Reference Classifier Lookup, and Nearest-Centroid Lookup realize the same registered endpoint and match the Teacher Endpoint. The clean equivalence supports a unified reference-to-endpoint interface and geometry-safe realization; it does not establish superiority over hard lookup.

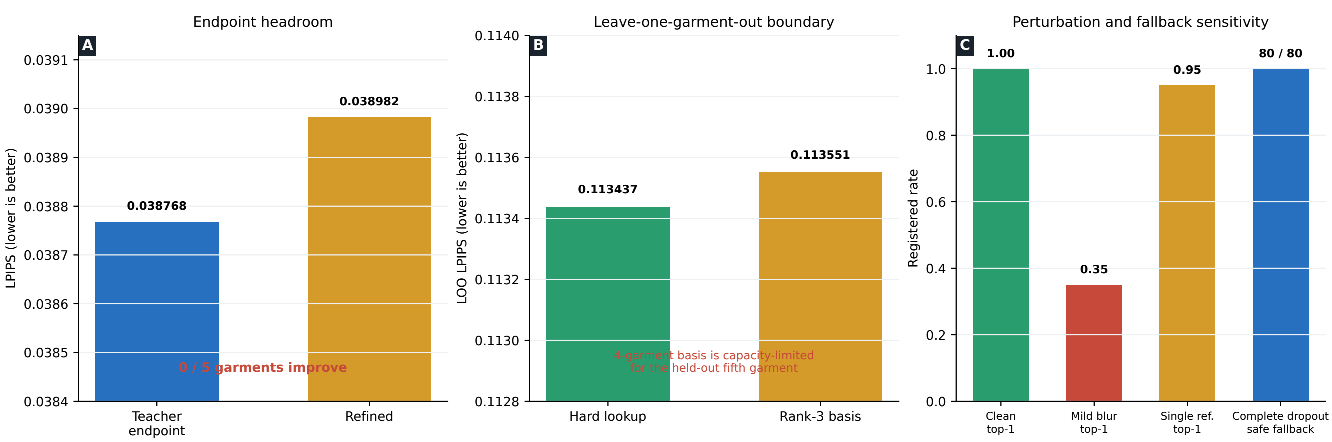

Figure 6: Endpoint-space analysis boundaries under the fixed-identity subject02, five-seen-garment closed-wardrobe protocol. (A) Test-time coefficient refinement does not improve Teacher Endpoint LPIPS (0.038982 versus 0.038768; 0/5 garments improve). (B) In leave-one-garment-out analysis, a four-garment rank-3 basis is capacity-limited for the held-out fifth garment and does not improve over hard lookup (0.113551 versus 0.113437 LPIPS). (C) Endpoint prediction is sensitive to mild blur and single-reference input, while complete reference dropout safely returns the Base Avatar in 80/80 cases. These are analysis boundaries, not deployment capabilities beyond the registered wardrobe.

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